

Mohammad Atif Quamar

MSc in Machine Learning at MBZUAI

✉ mohammad.atif@mbzuai.ac.ae
🌐 atifquamar.com
🔗 [atifquamar07](https://github.com/atifquamar07)
in [atifquamar](https://www.linkedin.com/in/atifquamar)
🔍 [Google Scholar](https://scholar.google.com/citations?user=atifquamar)

Research Interests

Pre-Training/Post-Training of LLMs, Mechanistic Interpretability, Machine Reasoning of LLMs, AI Alignment and Safety, Multi-Agent Systems

Education



M.Sc. in Machine Learning 2026 – 2028

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE

• Relevant Coursework: Machine Learning, Mathematical Foundations of Machine Learning, Principles of Applied Statistics, Deep Learning, Machine Learning with Python



B.Tech. in Computer Science and Biosciences 2020 – 2024

Indraprastha Institute of Information Technology (IIIT), Delhi, India

• Relevant Coursework: Machine Learning, Computer Vision, Natural Language Processing, Artificial Intelligence, Linear Algebra, Multivariate Calculus, Differential Equations

Publications

* *Equal contribution*

- [C1] **Reliable Chain-of-Thought via Prefix Consistency**
Naoto Iwase, Yuki Ichihara, Mohammad Atif Quamar and Junpei Komiyama
Advances in Neural Information Processing Systems (NeurIPS) 2026. [\[arXiv\]](#) [\[Page\]](#)
- [P1] **Self-Adapting Group of Experts for Multi-Agent Reasoning**
Mohammad Atif Quamar, Nurbek Tastan, Karthik Nandakumar, Junpei Komiyama
arXiv, 2026. Under Review. [\[arXiv\]](#) [\[Page\]](#)
- [P2] **Adaptive Blockwise Search: Inference-Time Alignment for Large Language Models**
Mohammad Atif Quamar, Mohammad Areeb*, Nishant Sharma, Ananth Shreekumar, Jonathan Rosenthal, Muslum Ozgur Ozmen, Mikhail Kuznetsov and Z. Berkay Celik*
arXiv, 2025. Under Review. [\[arXiv\]](#)
- [P3] **Privileged Solutions or Context-Induced Teacher Behaviour? Dissecting On-Policy Self Distillation**
Yuki Ichihara, Naoto Iwase, Mohammad Atif Quamar and Junpei Komiyama
arXiv, 2026. Under Review. [\[arXiv\]](#)
- [P4] **Prefix-Denoising Consistency for Diffusion Language Models**
Yuki Ichihara, Naoto Iwase, Mohammad Atif Quamar, Junpei Komiyama
arXiv, 2026. Under Review. [\[arXiv\]](#)
- [W1] **STARS: Synchronous Token Alignment for Robust Supervision in LLMs**
Mohammad Atif Quamar, Mohammad Areeb*, Mikhail Kuznetsov, Muslum Ozgur Ozmen, and Z. Berkay Celik*
Structured Probabilistic Inference & Generative Modeling Workshop at ICML 2026. [\[arXiv\]](#)

- [W2] **Logit-Entropy Adaptive Stopping Heuristic for Efficient Chain-of-Thought Reasoning**
Mohammad Atif Quamar and Mohammad Areeb
 Efficient Reasoning Workshop at NeurIPS 2025. [\[arXiv\]](#) [\[OpenReview\]](#)
- [W3] **Decoding Histone Modification Signatures of Non-Coding RNAs via FMs**
 Nishant Sharma, **Mohammad Atif Quamar** and Pengtao Xie
 Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences
 at NeurIPS 2025. [\[OpenReview\]](#)

Research Experience



Visiting Researcher Feb 2026 – Jul 2026
MBZUAI, Abu Dhabi, UAE Advisor: [Prof. Junpei Komiyama](#)
 • Working on methods to improve reasoning in language models through reinforcement learning, self-consistency, and test-time inference strategies, to enhance mathematical reasoning and coding performance while reducing unreliable reasoning paths. Publication - [\[P2\]](#).



Research Intern Sep 2025 – Feb 2026
University of Virginia, Virginia, United States Advisor: [Prof. Ferdinando Fioretto](#)
 • Developed a drift-resilient memory framework for code-execution agents using KL-constrained adapter updates, mitigating embedding distribution shifts during online learning to reduce unsafe code generation without compromising task success rates.



Research Intern Aug 2025 – Sep 2025
UC San Diego, California, United States Advisor: [Prof. Biwei Huang](#)
 • Proposed a multimodal Chain-of-Thought framework that interleaves text with latent visual representations in VLMs, with a two-stage SFT + RL training setup for modality switching, achieving strong gains across 11 multimodal reasoning benchmarks. Publication - [\[P3\]](#).



Research Intern Mar 2025 – Jul 2025
Purdue University, Indiana, United States Advisor: [Prof. Z. Berkay Celik](#)
 • Studied Bayesian sampling and inference-time alignment methods for language models, improving response quality and steering outputs toward harmlessness and positive sentiment. Publications - [\[C1\]](#) and [\[P1\]](#).

Industry Experience

Co-Founder Sep 2023 – Jan 2025
Insituate, New Delhi, India
 • Founded Insituate and grew it to \$250K ARR, building on-premise AI agents using open-source LLMs and RAG systems for enterprise use cases; deployments included Singapore Judiciary, Indian High Courts, Mizuho Bank, and PNC Bank.

Awards & Achievements

2026	Travel Grant. Received Travel Grant to attend Ellis Summer School in TU Munich, Germany.
2026	MBZUAI Scholarship. 100% tuition scholarship for M.Sc. in ML; \$100K scholarship package.
2024	NASSCOM GenAI Cohort. Selected among the top 15 AI startups in India.
2024	NVIDIA Inception. Secured \$150K worth of AWS credits.
2024	Microsoft Founder's Hub. Received \$150K worth of Azure credits.
2020	JEE Main. Ranked in the top 3% among 1.1 million candidates.
Present	Competitive Programming. 1800+ rated on CodeChef and LeetCode; world top 3%.